Nader Bou Hamdan

nader.bouhamdan@gmail.com • (747) 253-2707 • Los Angeles, California, USA • linkedin.com/in/naderbouhamdan

SUMMARY

A driven Mechatronics Engineer with a passion for Intelligent Systems and Technologies. Developed a comprehensive set of skills to create innovative and high-performance robotics and automation solutions. Seeking a role where I can leverage my technical skills and cooperative spirit to ensure successful completion of projects.

EDUCATION

Lebanese American University (LAU) – Byblos, Lebanon September 2020 – December 2024
Bachelor of Engineering, Major in Mechatronics Engineering

EXPERIENCE

Ascend Solutions – Research Intern
Remote, USA July 2024 – August 2024

- Conducted research on applying **Large Language Models** to medical topics, enhancing their ability to extract and analyze healthcare-related data.
- Fine-tuned and trained **3 Hugging Face models** as part of research efforts to improve medical insights.

Black Arrow Security and Systems – Engineering Intern
Doha, Qatar July 2024 – August 2024

- Developed and successfully closed a sale for an audiovisual ceremony for the **second-ranked** school in Qatar.
- Provided technical support for audiovisual setup and maintenance at multiple sites such as **Qatar Energy**.
- Attended biweekly meetings with the **Deputy general manager** to review and follow up on projects' progress.

BMB Smart – Robotics and Coding Instructor
Beirut, Lebanon July 2023 – December 2023

- Educated over **40 students**, ages 9–15, in developing innovation and teamwork skills through Lego Robotics, Scratch, and JavaScript.
- Led a summer camp with **30 students** in partnership with the **second-ranked** American university in Lebanon

ITP Media Group – Full Stack Web Developer Intern
Beirut, Lebanon June 2023 – August 2023

- Collaborated on the creation of an employee **task management portal**, developing the frontend using Angular.
- Optimized the portal by refactoring the SQL database and improving backend performance through **algorithm optimization** using Node.js and Express.js.

PROJECTS

Derma Cam Fall 2024

- Developed a **skin diagnosis** system using CNN and image analysis techniques.
- Utilized SolidWorks to design and prototype **camerabotics** for the system.
- Implemented and coded the AI and Computer Vision models with PyTorch and OpenCV and processed images using MATLAB.

Automatic Bartender, B.O.T.Y Spring 2024

- Engineered a conveyor belt with pump controls using **fuzzy logic** controllers implemented on MATLAB and Arduino.
- Designed a 3D-printed gripper on SolidWorks for a shaker to effectively mix beverages.

SKILLS AND HOBBIES

Languages: Arabic (Native), English (Fluent), and French (Basic)
Computer skills: Python (OpenCV, Sklearn, PyTorch), Java, Angular (TypeScript, HTML), SQL, AWS, Linux, Git, ROS, Docker, SolidWorks, MATLAB, LabView
Hobbies: Chess, Hiking, Gaming

CERTIFICATIONS AND INVOLVEMENTS

Administration Officer, LAU Simulation Models (MUN, etc.) – Beirut, Lebanon 2021 - 2024
Second Placement, Engineering Student Design Competition – Byblos, Lebanon 2023
Participation, LAU Case Competition – Beirut, Lebanon 2022
CSWA, SolidWorks 2021